

Correlation and Linear Regression

**Course: Foundations of Statistics for Data Analytics and
Machine Learning Using Excel**

@ DV Data Analytics, Bangalore

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Association between variables: Correlation

Association between Variables

- Association measures how two variables are related.
- Can be linear or non-linear.
- Examples: height vs. weight, temperature vs. ice cream sales.

Types of Association:

- **Positive Association:** Increase in one variable leads to an increase in another.
- **Negative Association:** Increase in one variable leads to a decrease in another.
- **No Association:** No apparent relationship between variables.

Quantifying Association

- **Covariance:** Indicates direction of relationship but not strength.
 - It is a measure of how the two variables change together.
- **Correlation Coefficient (r):** Measures linear relationship.
 - $r > 0$: Positive correlation
 - $r < 0$: Negative correlation
 - $r = 0$: No correlation
- **Example: Correlation Calculation**
 - Given: $X = \{1, 2, 3, 4, 5\}$, $Y = \{2, 4, 6, 8, 10\}$
 - Compute correlation coefficient r .
 - Since $Y = 2X$, we get $r = 1$ (perfect positive correlation).

Examples

Example 1: Height vs. Weight

- Typically, as height increases, weight also increases.
- Positive correlation is expected.
- Helps in BMI calculations and health studies.

Example 2: Temperature vs. Ice Cream Sales

- Higher temperatures lead to increased ice cream sales.
- Strong positive correlation exists.
- Useful in demand forecasting.

Example 3: Study Hours vs. Exam Scores

- More study hours often lead to higher exam scores.
- Positive correlation exists, but outliers may affect results.
- Important in education analytics.

Why is it Important

- Understanding relationships between variables is crucial in data science.
- Correlation and covariance help measure these relationships.
- Used in feature selection, PCA, regression, etc.

Covariance

- Measures how two random variables change together.
- Formula:

$$\text{Cov}(X, Y) = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

- Positive covariance: variables increase together.
- Negative covariance: one increases while the other decreases.

$X_i - \bar{X} \parallel Y_i - \bar{Y} \parallel (X_i - \bar{X}) \times (Y_i - \bar{Y})$		
+ve	+	+ve
+ve	-	-ve
-ve	+	-ve
-ve	-	+ve

- Observe Association for different data.
 - Positive Association.
 - Negative Association.
 - No Association.

Correlation

- Standardized version of covariance.
- Pearson correlation coefficient formula:

$$r = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \quad (1)$$

- Ranges from -1 to 1.
- +1: Strong positive relationship.
- -1: Strong negative relationship.
- 0: No linear relationship.

Covariance vs. Correlation

- Covariance is in original units, while correlation is dimensionless.
- Correlation is easier to interpret.
- Correlation is used for comparing different datasets.

Why It Matters in Data Science?

- Feature selection: Remove redundant features.
- Principal Component Analysis (PCA): Uses covariance matrix.
- Regression analysis: Identifies relationships between variables.
- Portfolio management: Helps in risk assessment.

Example Case Study: Stock Prices

- Covariance helps analyze stock price movements.
- Correlation shows market trends.
- Used in risk assessment and portfolio optimization.

Visualizing Covariance

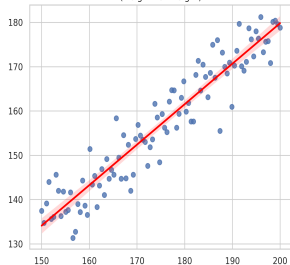
- Scatter plots help visualize relationships.
- Heatmaps of covariance matrices show variable interactions.

Visualizing Correlation

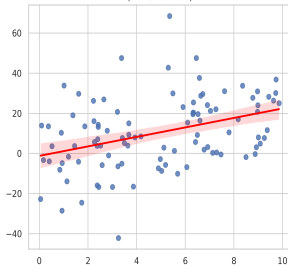
- Correlation matrix helps identify highly correlated variables.
- Useful in dimensionality reduction techniques.
- Heatmaps.

Scatter Plots

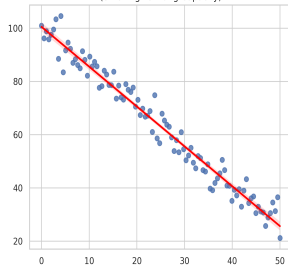
Strong Positive Correlation
(Height vs. Weight)



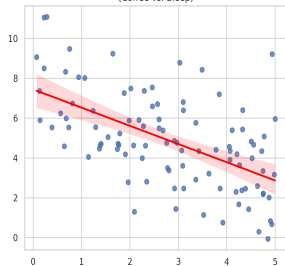
Weak Positive Correlation
(TV vs. Grades)



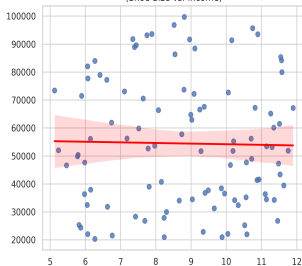
Strong Negative Correlation
(Smoking vs. Lung Capacity)



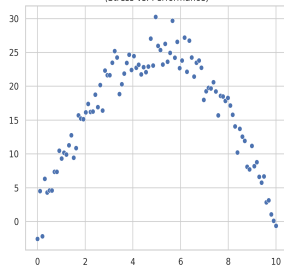
Weak Negative Correlation
(Coffee vs. Sleep)



No Correlation
(Shoe Size vs. Income)



Non-linear Correlation
(Stress vs. Performance)



Common Misinterpretations

- Correlation does not imply causation.
- Outliers can affect correlation values.
- Nonlinear relationships may exist despite low correlation.

Summary

- Covariance measures joint variability but lacks interpretability.
- Correlation standardizes covariance and is easier to interpret.
- Both are fundamental in statistical analysis and machine learning.

Linear Regression

Linear Regression

- Correlation measures the strength and direction of a linear relationship between two variables.
- Strong correlation suggests a potential linear relationship useful for regression.
- Correlation indicates whether a linear relationship exists but does not imply causation.
- Regression builds on correlation by modeling one variable (y) as a function of another (x):

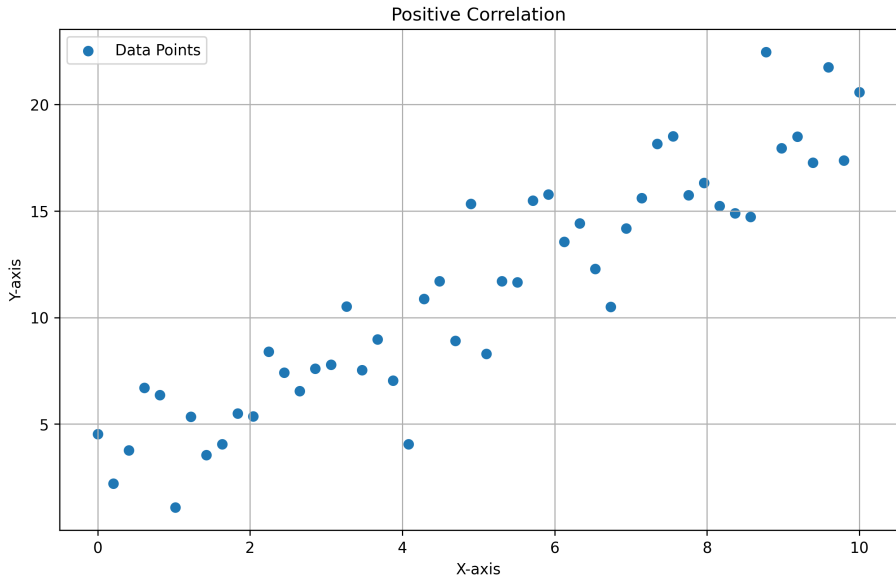
$$y = \beta_0 + \beta_1 x$$

- A strong correlation ($|r|$ close to 1) suggests that linear regression might be a good fit.

Building a Regression Model

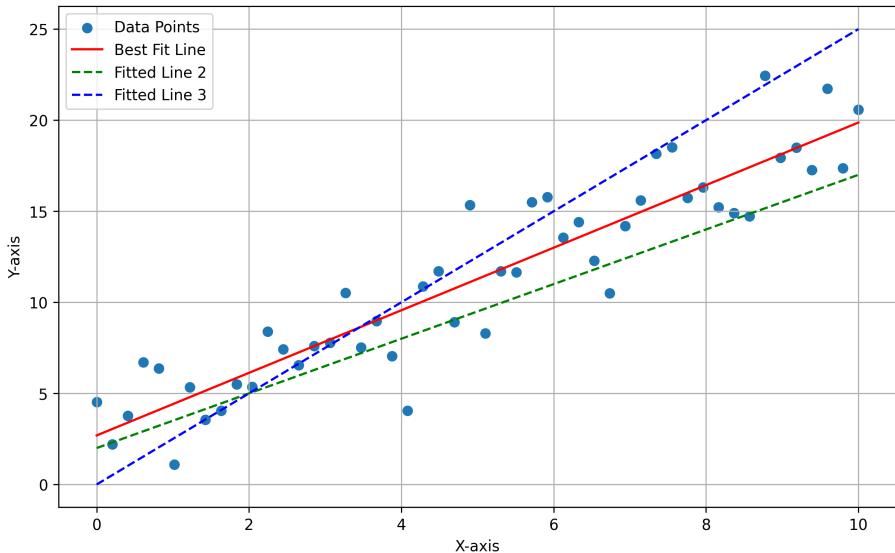
- Steps to build a regression model:
 - ① Check correlation to identify potential predictors.
 - ② Fit a linear model and estimate parameters (β_0, β_1).
 - ③ Evaluate goodness-of-fit (e.g., R^2 , residual analysis).
 - ④ Interpret the model and validate assumptions (linearity, normality, homoscedasticity).
- Even with high correlation, other factors (e.g., omitted variables, multicollinearity) can affect model reliability.

Linear Relationship



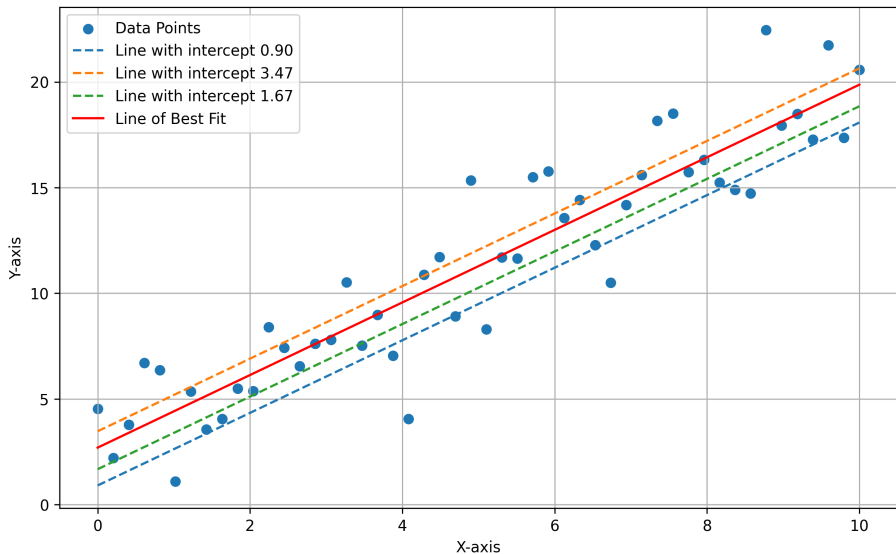
Best Fit Line

Positive Correlation with Fitted Lines

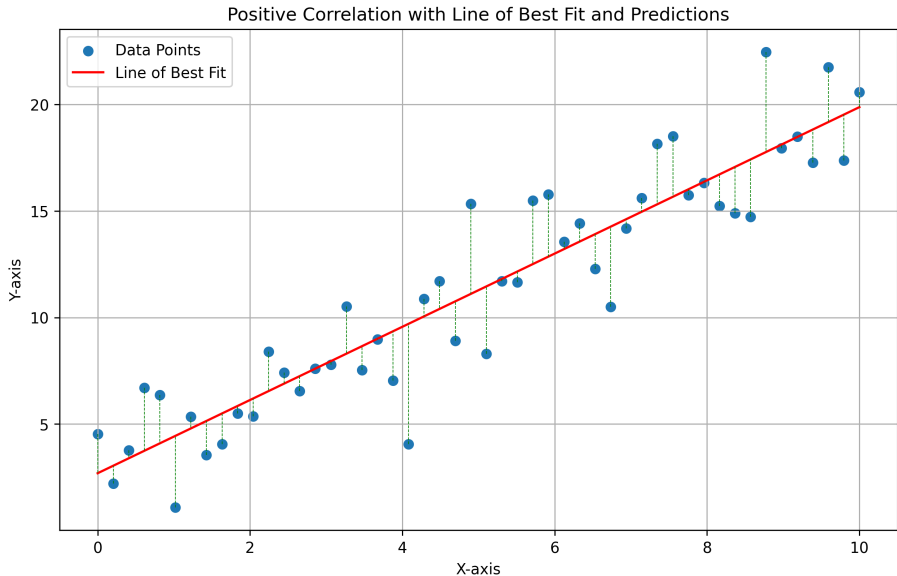


Best Fit Line

Positive Correlation with Different Fits

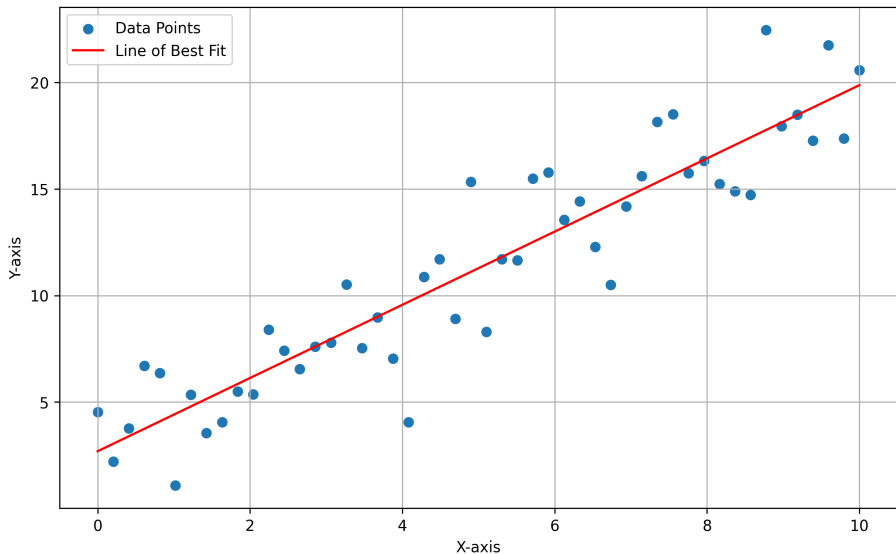


how is the best Fit Line Chosen



Best Fit Line

Positive Correlation with Line of Best Fit



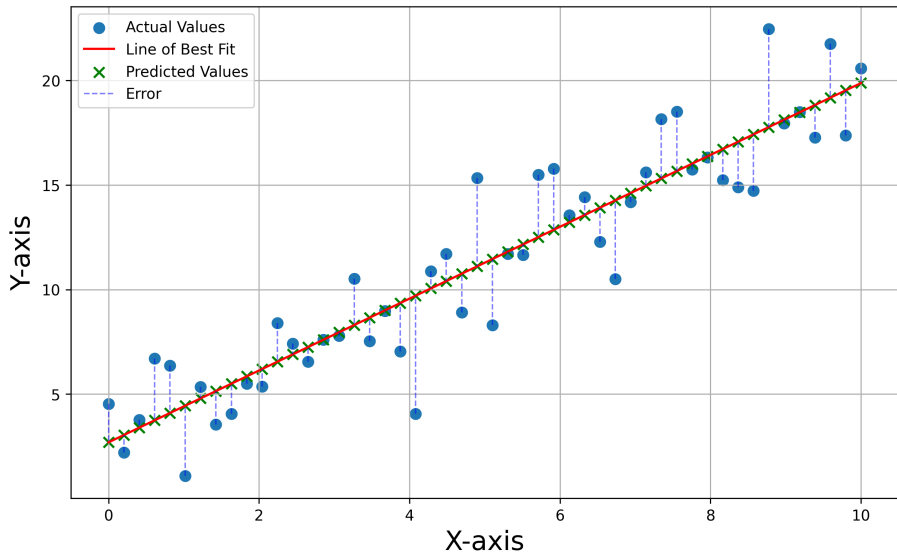
Definition

- Linear regression is a statistical method used to model the relationship between a dependent variable (Y) and one or more independent variables (X).

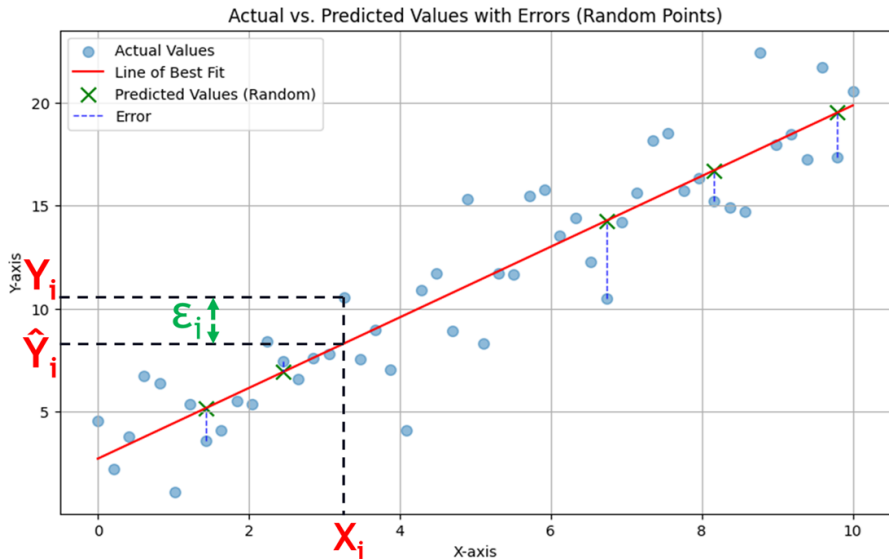
$$Y = \beta_0 + \beta_1 X + \varepsilon$$

Linear Regression

Actual vs. Predicted Values with Errors



Prediction with Linear Regression



Linear Regression

Linear Regression

- It is constructing the regression line.
- The regression line gives an estimate of the actual value.

We Know that

$$Y = \hat{Y} + \varepsilon$$

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

Where

$$\hat{Y} = \beta_0 + \beta_1 X$$

Regression model building

- Simple Regression is construction the line $\hat{Y}$ from X and Y .
- It means determining the coefficients β_0 and β_1 .

Types of Linear Regression

- Simple Regression: One predictor variable/Feature.
- Multiple Regression: More than one predictor variable.

Model Building

Model Building in either case refers to determining the coefficients.

What is R^2 ?

- R^2 , or the **coefficient of determination**, measures how well the regression line explains the variability of the dependent variable.
- Ranges from **0 to 1**.
- A higher R^2 indicates a better fit of the model.

Logistic Regression

What is Logistic Regression?

- A classification algorithm used when the dependent variable is binary (0 or 1).
- Models the probability that an instance belongs to a class.
- Uses the logistic (sigmoid) function to squash output between 0 and 1.

Logistic vs Linear Regression

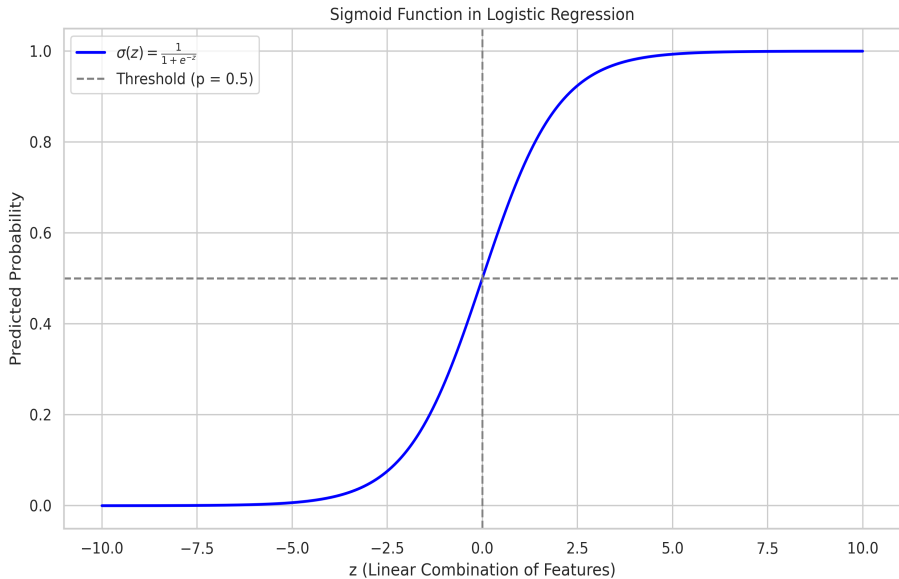
- **Linear Regression:** Predicts continuous outcomes.
- **Logistic Regression:** Predicts probabilities, then classifies.
- **Problem:** Linear model can produce values < 0 or > 1 .
- **Solution:** Apply sigmoid function.

The Logistic (Sigmoid) Function

$$\sigma(z) = \frac{1}{1 + e^{-z}} \quad \text{where } z = \beta_0 + \beta_1 x$$

- Converts linear model output into a probability.
- Output: between 0 and 1.

Visual Understanding



Applications of Logistic Regression

- Medical diagnosis (e.g., disease presence)
- Marketing (e.g., customer churn)
- Fraud detection
- Credit scoring

Summary

- Logistic regression is a probabilistic classifier.
- Uses sigmoid function to map predictions to $[0, 1]$.
- Coefficients interpreted in terms of log-odds and odds ratios.
- Evaluated using classification metrics.

Time Series Forecasting

What is a Time Series?

- A sequence of data points indexed in time order.
- It is an ordered sequence of values of a variable at equally spaced time intervals.
- Examples:
 - Daily stock prices
 - Monthly sales revenue
 - Annual rainfall
- Formally: y_t , where $t = 1, 2, \dots, T$

Cross-Sectional vs Time Series Data

Cross-Sectional Data

- Collected at a single point in time
- Observations across different entities
- No temporal order

Example:

- Income levels of 100 individuals in 2024
- Sales from 50 stores on Jan 1st

Time Series Data

- Collected over multiple time periods
- Ordered chronologically
- Temporal dependence is key

Example:

- Monthly unemployment rate (2010–2024)
- Daily temperature readings

What Does Temporal Mean in Time Series?

- **Temporal** = Related to time
- In time series, observations are **ordered in time**
- Each value may depend on **previous values**
- **Time order matters** — you can't shuffle the data!

Example: Daily Temperatures Ordered (Temporal)

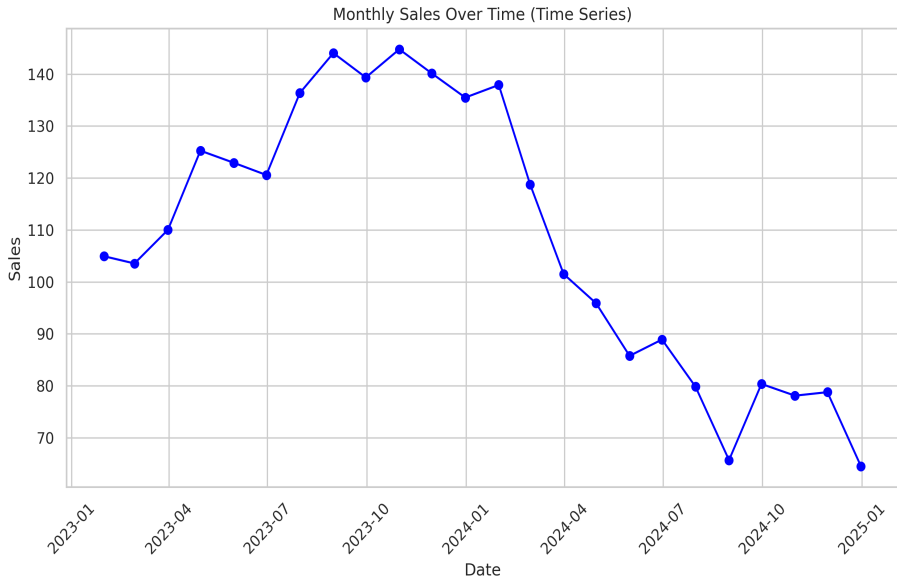
- Mon: 20°C
- Tue: 21°C
- Wed: 23°C
- Thu: 22°C

Shuffled (Not Temporal)

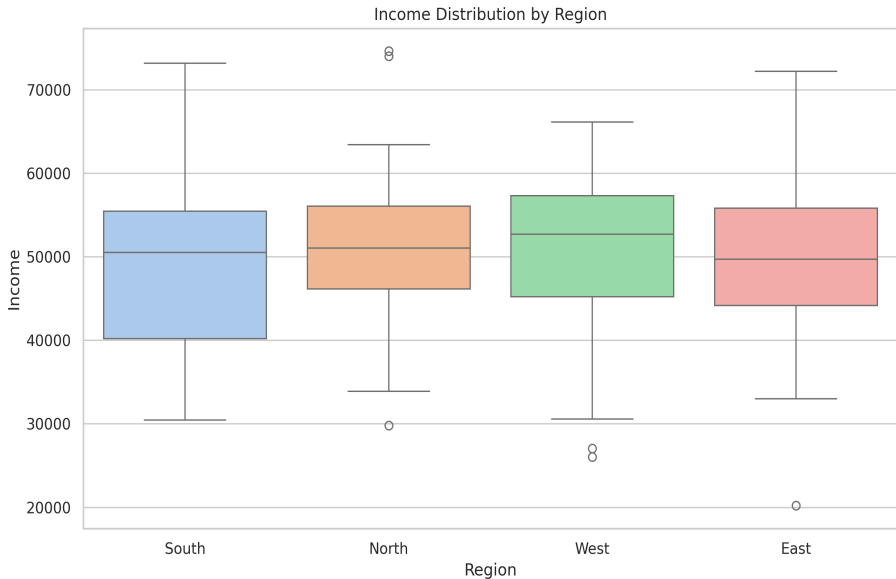
- Wed: 23°C
- Mon: 20°C
- Thu: 22°C
- Tue: 21°C

Losing the time order means losing the context.

Time Series vs Cross-Sectional Data



Time Series vs Cross-Sectional Data



Introduction to Time Series Forecasting

- Forecasting involves using past time-dependent data to predict future values.
- It captures patterns such as:
 - **Trend** (long-term increase or decrease)
 - **Seasonality** (repeating patterns at fixed intervals)
 - **Noise** (random variation)
- Time series forecasting is crucial in business, finance, weather, and more.

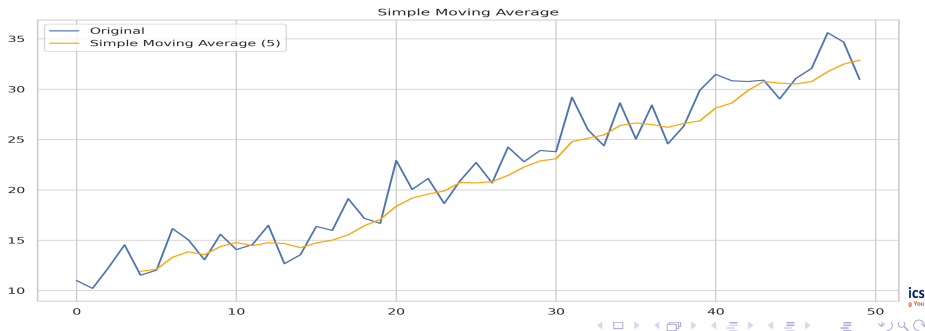
Forecasting assumes that the future is, at least partially, dependent on the past.

Simple Moving Average (SMA)

- The forecast is the average of the last n observations:

$$\hat{y}_t = \frac{1}{n} \sum_{i=1}^n y_{t-i}$$

- Smooths out short-term fluctuations.
- Best suited for series with no clear trend or seasonality.

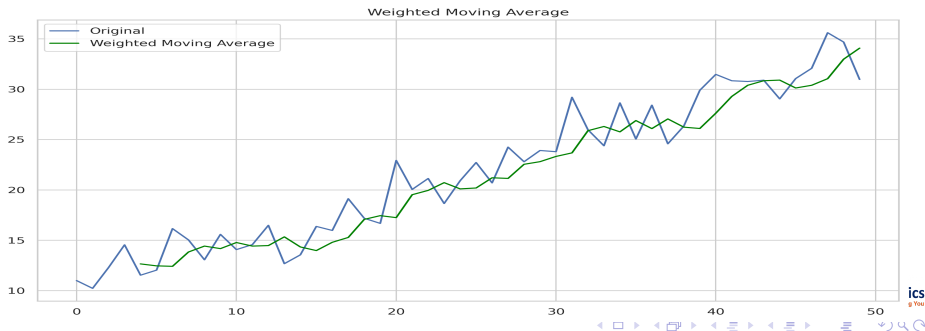


Weighted Moving Average (WMA)

- Assigns more weight to recent observations:

$$\hat{y}_t = \sum_{i=1}^n w_i y_{t-i}, \quad \text{where } \sum w_i = 1$$

- More responsive to recent changes in the data.
- Useful when recent data points are more relevant.

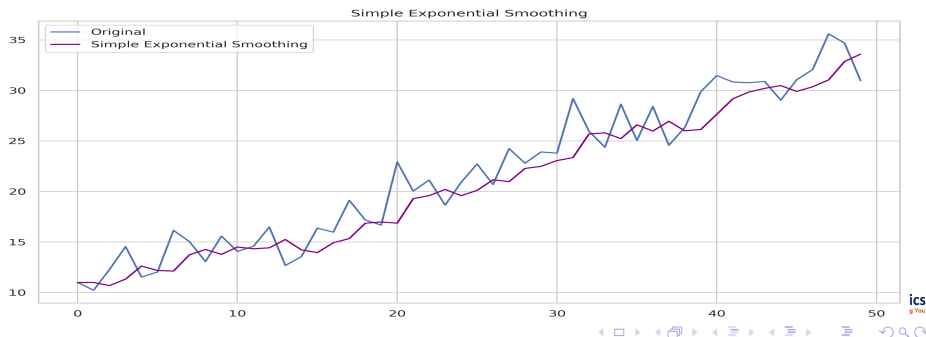


Simple Exponential Smoothing (SES)

- Forecast is a weighted average of past observations with exponentially decreasing weights:

$$\hat{y}_{t+1} = \alpha y_t + (1 - \alpha) \hat{y}_t, \quad 0 < \alpha < 1$$

- Good for stationary series (no trend/seasonality).
- More flexible than moving averages due to the smoothing factor α .



Holt's Linear Trend Method

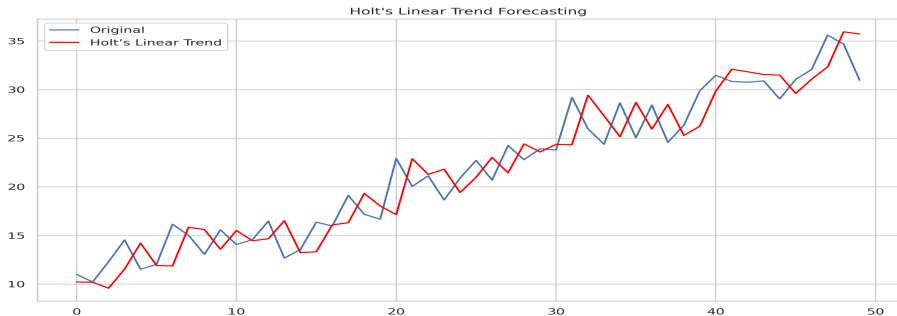
- Extends SES to capture linear trend:

$$l_t = \alpha y_t + (1 - \alpha)(l_{t-1} + b_{t-1})$$

$$b_t = \beta(l_t - l_{t-1}) + (1 - \beta)b_{t-1}$$

$$\hat{y}_{t+h} = l_t + hb_t$$

- l_t : level, b_t : trend, α , β : smoothing parameters.
- Suitable for data with a trend but no seasonality.



Real-World Applications

- **Finance:** Stock market prediction
- **Retail:** Inventory planning
- **Meteorology:** Weather forecasting
- **Energy:** Load forecasting

Concluding Remarks

Concluding Remarks

- Statistics provides the foundation for understanding data.
- It helps us make informed decisions based on data analysis.
- Essential for interpreting results and improving machine learning models.

Understanding Data

- Data comes with variability and uncertainty.
- Statistics helps to quantify uncertainty and variability.
- Concepts like mean, variance, and standard deviation describe the data's distribution.

Hypothesis Testing

- In machine learning, we often test the significance of a model or a feature.
- Hypothesis testing helps determine if the findings are meaningful or due to random chance.
- Statistical tests guide model selection and evaluation.

Model Evaluation

- Statistics is essential in measuring model performance.
- Key metrics: Accuracy, Precision, Recall, F1-Score, ROC-AUC, etc.
- Statistical methods help assess whether a model generalizes well to unseen data.

Conclusion

- Statistics is foundational to machine learning.
- It aids in data analysis, model evaluation, and improving prediction accuracy.
- A strong understanding of statistics enhances the ability to build effective machine learning models.

THANK YOU